

crude oil. There are a few exceptions to this rule. Physical gold and other precious metals can now be traded as an ETF.

Most people trade commodity futures rather than physical commodities. These are paper contracts that grant ownership rights some time in the future. Futures contracts are based on a future delivery price rather than the current spot price. The contracts can be traded, like stocks, on a futures exchange.

Commodity futures represent an obligation to buy commodities in the months ahead based on a price that is negotiated today. A contract is a standardized agreement to buy or sell a specific commodity type and quantity at that future date and price. Only a small amount of money is needed to secure a futures contract, and this gives investors a huge amount of leverage.

Calculating Futures Prices

Futures prices can differ significantly from spot prices. Calculating the futures price begins with the current spot price, and then it adjusts for a variety of factors that may include seasonal changes in supply and demand, borrowing costs, storage expenses, and other costs-of-carry expenses. If futures prices are trading at a higher level than the current spot price, it is known as *contango*. When futures prices are trading lower than spot prices, it is known as *backwardation*.

[Figure 10-3](#) is an example of the oil futures market in contango and backwardation. It is based on a \$60-per-barrel expected spot price for oil.

There has been a long debate about the returns on commodities futures. Some studies suggest that the returns are comparable to stocks even though the long-term spot market has not achieved those results. Other academic studies show that futures returns tend to be closer to the inflation rate. The discrepancies revolve

FIGURE 10-3

Oil Futures Prices in Backwardation and Contango